WORKED SOLUTION

# STEP III 2004, Question 12

Probability and Statistics • Indicators and Derangements

**STEP III 2004, Question 12**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q12

## The key idea

Writing a count as a sum of indicator variables is one of the most useful tricks in probability. Expectations of indicators are just probabilities, so $\mathrm{E}(N)$ comes for free; the variance needs the covariances between pairs, and these are small but not zero, because the shirts are not assigned independently. Once we know $N$ has mean and variance both equal to 1, the Poisson distribution with parameter 1 is the natural model to try.

## Solution

### The single indicator $C_1$

Every one of the $m!$ assignments of shirts is equally likely, so shirt 1 is equally likely to go to any of the $m$ players. Hence $\mathrm{P}(C_1 = 1) = \frac{1}{m}$, and since $C_1$ takes only the values 0 and 1,

$$\mathrm{E}(C_1) = 1 \cdot \mathrm{P}(C_1 = 1) = \frac{1}{m},$$

as required. Because $C_1^2 = C_1$ (as $0^2 = 0$ and $1^2 = 1$), we also have $\mathrm{E}(C_1^2) = \frac{1}{m}$, so

$$\mathrm{Var}(C_1) = \frac{1}{m} - \frac{1}{m^2} = \frac{m-1}{m^2}.$$

### The covariance of $C_1$ and $C_2$

Here we need $m \geq 2$. The product $C_1C_2$ equals 1 exactly when players 1 and 2 both have their own shirts, and 0 otherwise. Given that player 1 has shirt 1, the remaining $m-1$ shirts are shared out at random among the other $m-1$ players, so player 2 gets shirt 2 with probability $\frac{1}{m-1}$. Therefore

$$\mathrm{E}(C_1C_2) = \mathrm{P}(C_1 = 1,\, C_2 = 1) = \frac{1}{m} \cdot \frac{1}{m-1},$$

and

$$\mathrm{Cov}(C_1, C_2) = \mathrm{E}(C_1C_2) - \mathrm{E}(C_1)\,\mathrm{E}(C_2) = \frac{1}{m(m-1)} - \frac{1}{m^2}$$
$$= \frac{m-(m-1)}{m^2(m-1)} = \frac{1}{m^2(m-1)}.$$

It is positive: knowing that player 1 has the right shirt removes one wrong shirt from the pool for player 2, which slightly raises their chance.

**Ask yourself** Why isn't the covariance zero? Aren't the players' shirts chosen separately?

They are not. Shirts are assigned one to each player, so what player 1 gets affects what is left for everyone else. The $C_i$ are *dependent*, and that is exactly why the covariance terms must be included in $\mathrm{Var}(N)$.

#### The mean and variance of $N$

By symmetry, every $C_i$ has the same distribution as $C_1$, and every pair $C_i, C_j$ with $i \neq j$ has the same joint distribution as $C_1, C_2$ (relabelling players and shirts together doesn't change the random assignment). By linearity of expectation, which needs no independence,

$$\mathrm{E}(N) = \sum_{i=1}^{m} \mathrm{E}(C_i) = m \cdot \frac{1}{m} = 1.$$

For the variance of a sum we add all the variances and all the covariances over ordered pairs $i \neq j$, of which there are $m(m-1)$:

$$\mathrm{Var}(N) = \sum_{i=1}^{m} \mathrm{Var}(C_i) + \sum_{i \neq j} \mathrm{Cov}(C_i, C_j)$$
$$= m \cdot \frac{m-1}{m^2} + m(m-1) \cdot \frac{1}{m^2(m-1)}$$
$$= \frac{m-1}{m} + \frac{1}{m} = 1.$$

So $\mathrm{E}(N) = \mathrm{Var}(N) = 1$, as required. (For $m = 1$ we would have $N = 1$ always, so $\mathrm{Var}(N) = 0$; the question implicitly assumes $m \geq 2$, as the covariance already required.)

#### Which approximation?

A Normal approximation is poor whatever the value of $m$. $N$ takes only the whole-number values $0, 1, 2, \ldots$, and with mean 1 and standard deviation 1 almost all its probability sits on the three or four values $0, 1, 2, 3$. The distribution is cut off at 0, just one standard deviation below the mean, so it is strongly skewed to the right; a $\mathrm{N}(1, 1)$ distribution is symmetric and puts noticeable probability (about 7% even with a continuity correction) below $-\frac{1}{2}$, where $N$ can never be. Crucially, increasing $m$ doesn't

help: the mean and variance stay at 1 for every $m$, so the shape never spreads out into a bell.

A Poisson approximation, on the other hand, is plausible. $N$ counts the "successes" among a large number $m$ of events, each with a small probability $\frac{1}{m}$, and these events are only weakly dependent (the covariance $\frac{1}{m^2(m-1)}$ is tiny). That is the setting in which counts are approximately Poisson. Moreover, a Poisson variable has its mean equal to its variance, and $N$ has exactly this property, so $\mathrm{Po}(1)$ is the natural choice.

#### The case $m = 4$

There are $4! = 24$ equally likely assignments. Writing an assignment as the list of shirts worn by players $1, 2, 3, 4$, the ones in which nobody has their own shirt are

$$\begin{array}{llllll} 2143, & 2341, & 2413, & 3142, & 3412, \\ 3421, & 4123, & 4312, & 4321. & & \end{array}$$

(To list them systematically: player 1 has shirt 2, 3 or 4, and in each case there are exactly three ways to finish, so $3 \times 3 = 9$.) Hence

$$\mathrm{P}(N = 0) = \frac{9}{24} = \frac{3}{8} = 0.375.$$

The Poisson approximation with parameter 1 gives $\mathrm{P}(N = 0) \approx \mathrm{e}^{-1}$. Using $\mathrm{e} \approx 2.72$,

$$\mathrm{e}^{-1} \approx \frac{100}{272} = \frac{25}{68}.$$

The relative error, measured against the true value, is

$$\frac{\frac{3}{8} - \frac{25}{68}}{\frac{3}{8}} = 1 - \frac{25}{68} \cdot \frac{8}{3} = 1 - \frac{200}{204} = \frac{1}{51} \approx 0.0196,$$

which is about 2%, as claimed. (If instead we measure the error relative to the approximation, we get $\frac{1}{50}$, exactly 2%; either way the agreement is remarkably good for such a small $m$.)

## Checks

- For $m = 2$ the only possibilities are both right ($N = 2$) or both wrong ($N = 0$), each with probability $\frac{1}{2}$: mean 1, variance 1. Good.
- A computer enumerating all permutations for $m = 2, \ldots, 7$ confirmed $\mathrm{E}(C_1) = \frac{1}{m}$, $\mathrm{Var}(C_1) = \frac{m-1}{m^2}$, $\mathrm{Cov}(C_1, C_2) = \frac{1}{m^2(m-1)}$ and $\mathrm{E}(N) = \mathrm{Var}(N) = 1$ exactly, and found 9 derangements of 4 objects, so $\mathrm{P}(N = 0) = 0.375$ against $1/2.72 \approx 0.3676$.
- Note that $N = m - 1$ is impossible: if all but one player have the right shirt, so does the last. This is another sign that $N$ is very far from Normal.

## Common slips

- Treating the $C_i$ as independent, which gives $\mathrm{Var}(N) = \frac{m-1}{m}$ instead of 1.
- Counting only $\binom{m}{2}$ covariance terms instead of $m(m-1)$, i.e. forgetting the factor 2 in $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)$.
- Listing the derangements carelessly: a systematic count ($3 \times 3$) avoids missing one.

## Beyond STEP

This is the classical "matching" or derangement problem. Using inclusion–exclusion one can show that $\mathrm{P}(N=0) = \sum_{k=0}^{m} \frac{(-1)^k}{k!}$, which tends to $\mathrm{e}^{-1}$ extremely fast, and in fact $N$ converges in distribution to $\mathrm{Po}(1)$ as $m \to \infty$. The broader principle, that a count of many rare, weakly dependent events is nearly Poisson, is made precise by the Chen–Stein method. Blitzstein and Hwang, *Introduction to Probability*, uses indicator variables and the matching problem as recurring examples.

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